

Where Limit Work Goes Wrong

Answer Key

AP Calculus AB/BC

Quick Answers

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|---|---|
| 1. (a) the limit = $\frac{15}{4}$, — | 5. (a) the correct value = $-\infty$, — |
| 2. (a) the correct value = 4, — | 6. (a) from the left = $-\infty$, (b) from the right = |
| 3. (a) from the left = $-\infty$, (b) from the right = | $-\infty$ |
| ∞ | 7. (a) the correct value = ∞ , — |
| 4. $-\frac{1}{2}$ | 8. left-hand limit = -2 , right-hand limit = 2, — |
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What is verified

11 of 16 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problems 1, 2, 5, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

LIM-1 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 16 tagged checks.

Common wrong answers

2. If they got 0: reported the indeterminate form 0/0 as the answer 0
8. If they got 2: dropped the absolute value bars and cancelled as if the denominator were $x - 1$ on both sides
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1. (a) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 + 2x}{x + 1}$. (b) How can you tell in advance that substitution is safe?

Solution (a): The denominator at $x = 3$ is $3 + 1 = 4 \neq 0$, so the quotient is defined at 3 and the function is continuous there. Substitution is therefore legal:

$$\frac{3^2 + 2(3)}{3 + 1} = \frac{9 + 6}{4} = \frac{15}{4}. \quad \boxed{\frac{15}{4}}$$

Solution (b) — instructor-judged, no machine check. A correct answer says the denominator is not zero at the target: $x + 1 = 4$ at $x = 3$, so the expression is defined there and the limit equals the value. Full credit names the non-zero denominator explicitly.

Do not accept “because it is a polynomial” on its own — the expression is a quotient, and checking the bottom is exactly the habit this problem is building.

- 2.** Find and fix. The student wrote $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = \frac{0}{0} = 0$. (a) Correct value. (b) Name the error.

Solution (a): Substitution genuinely does give $\frac{0}{0}$, which means it has failed — not that the answer is 0. Factor and cancel instead:

$$\frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2} = x+2 \quad (x \neq 2),$$

so the limit is $2+2$.

4

Solution (b) — instructor-judged, no machine check. The diagnosis must say that $\frac{0}{0}$ is an *indeterminate form*, not a value: it carries no information about the limit and simply signals that the expression must be rewritten first. Full credit also names the skipped step — factor the numerator as $(x-2)(x+2)$ and cancel the common factor. Accept “you cannot divide zero by zero” only when it is paired with what to do instead.

- 3.** $f(x) = \frac{1}{x-3}$. Find (a) $\lim_{x \rightarrow 3^-} f(x)$ and (b) $\lim_{x \rightarrow 3^+} f(x)$.

Solution: The numerator is the constant $1 > 0$, so the direction is decided entirely by the sign of $x-3$.

For x just below 3, $x-3$ is a small *negative*, so 1 divided by it is a large negative number.

$-\infty$

For x just above 3, $x-3$ is a small *positive*, so the quotient is a large positive number.

$+\infty$

Because the two sides run in opposite directions, the two-sided limit does not exist.

- 4.** Evaluate $\lim_{x \rightarrow -1} \frac{x^2+3x+2}{x^2-1}$.

Solution: At $x = -1$ both parts are zero, so factor both completely:

$$\frac{x^2+3x+2}{x^2-1} = \frac{(x+1)(x+2)}{(x+1)(x-1)} = \frac{x+2}{x-1} \quad (x \neq -1).$$

Substituting $x = -1$ into the reduced form gives $\frac{-1+2}{-1-1} = \frac{1}{-2}$.

$-\frac{1}{2}$

Cancelling the x^2 terms instead — a subtraction is not a factor — would have produced 1, which is the trap this problem is built around.

- 5.** Find and fix. The student answered $+\infty$ for $\lim_{x \rightarrow 2^-} \frac{x+4}{x-2}$. (a) Correct value. (b) Name the error.

Solution (a): Approaching 2 from the left means $x < 2$, so $x-2$ is a small *negative* number, while the numerator is near $+6$. A positive divided by a small negative is a large negative:

$-\infty$

Solution (b) — instructor-judged, no machine check. The student checked the numerator's sign and stopped. The diagnosis must say that the *denominator's* sign was never checked: to the left of 2 the quantity $x - 2$ is negative, so positive over negative gives $-\infty$. Full credit states that the numerator's sign alone cannot decide the direction. Do not accept "they forgot a minus sign" with no reference to which quantity is negative.

6. $g(x) = \frac{2-x}{(x-5)^2}$. Find (a) $\lim_{x \rightarrow 5^-} g(x)$ and (b) $\lim_{x \rightarrow 5^+} g(x)$.

Solution: A squared denominator is positive on *both* sides — that is what makes this different from problem 3. The numerator tends to $2 - 5 = -3 < 0$.

From the left: negative over a small positive, so

$-\infty$

From the right: the same negative over the same small positive, so

$-\infty$

Here the two sides agree, so it is correct to write $\lim_{x \rightarrow 5} g(x) = -\infty$.

7. Find and fix. The student answered $-\infty$ for $\lim_{x \rightarrow -2^-} \frac{x-6}{x+2}$. (a) Correct value. (b) Name the error.

Solution (a): Near $x = -2$ the numerator is close to -8 , so it is negative — that much the student had right. But approaching from the left means $x < -2$, so $x + 2$ is a small *negative* too. A negative divided by a negative is positive:

$+\infty$

Solution (b) — instructor-judged, no machine check. The diagnosis must say that both signs matter, and that the denominator's sign was not checked: on the left of -2 the quantity $x + 2$ is negative, so negative-over-negative gives $+\infty$. Full credit states the negative-over-negative reasoning. Award partial credit for correcting the answer with no reason given.

8. Challenge — find and fix. The student cancelled $|x - 1|$ as $x - 1$ and answered 2 for $\lim_{x \rightarrow 1} \frac{x^2-1}{|x-1|}$. (a) Both one-sided limits. (b) Name the two faults.

Solution (a): Split the absolute value by side.

For $x < 1$: $|x - 1| = -(x - 1)$, so $\frac{(x-1)(x+1)}{-(x-1)} = -(x+1) \rightarrow -(1+1)$, giving

-2

For $x > 1$: $|x - 1| = x - 1$, so $\frac{(x-1)(x+1)}{x-1} = x+1 \rightarrow 1+1$, giving

2

Solution (b) — instructor-judged, no machine check. Two faults must be named. **First**, the absolute value was dropped: for $x < 1$ the denominator is $1 - x$, not $x - 1$, so the cancellation as written is valid only on the right-hand side. **Second**, the work never separated the two sides at all — and since the one-sided values are -2 and 2 , the two-sided limit does not exist, so reporting any single number is wrong in form as well as in value. Full credit names both faults; award partial credit for either one alone.